

Recitation 1

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Outline

1. **Review**

2. **Recitation**

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Review

- Sample space
 - List of possible outcomes
 - Mutually exclusive & Collectively exhaustive

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- Event
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 - Subset of the sample space
 - A collection of possible outcomes
- Axioms of probability
 - Nonnegativity: $\mathbf{P}(A) \geq 0$
 - Normalization: $\mathbf{P}(\text{universe}) = 1$
 - Countable additivity:
 - If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$
 - If $A_1, A_2, \dots$ are disjoint events, then
$$\mathbf{P}(A_1 \cup A_2 \cup \dots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots$$

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1. Give a mathematical derivation of the formula

$$\mathbf{P}((A \cap B^c) \cup (A^c \cap B)) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$$

Your derivation should be a sequence of steps, with each step justified by appealing to one of the probability axioms.

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$$\mathbf{P}((A \cap B^c) \cup (A^c \cap B)) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$$

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Solution:

Additivity:

$$\mathbf{P}(A \cap B^c) = \mathbf{P}(A) - \mathbf{P}(A \cap B)$$

$$\mathbf{P}(A^c \cap B) = \mathbf{P}(B) - \mathbf{P}(A \cap B)$$

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2. Out of the students in a class, 60% are geniuses, 70% love chocolate, and 40% fall into both categories. Determine the probability that a randomly selected student is neither a genius nor a chocolate lover.

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Solution:

Let

A : The event that the randomly selected student is a genius.

B : The event that the randomly selected student loves chocolate.

We have

$$\begin{aligned} 1 &= \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B) + \mathbf{P}(A^c \cap B^c) \\ &= 0.6 + 0.7 - 0.4 + \mathbf{P}(A^c \cap B^c). \end{aligned}$$

3. A six-sided die is loaded in a way that each even face is twice as likely as each odd face. Construct a probabilistic model for a single roll of this die, and find the probability that a 1, 2, or 3 will come up.

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Solution:

Let c denote the probability of a single odd face. Then the probability of a single even face is $2c$, and by adding the probabilities of the 3 odd faces and the 3 even faces, we get $9c = 1$. Thus, $c = 1/9$. The desired probability is

$$\mathbf{P}(\{1, 2, 3\}) = 4c = 4/9.$$

4. Romeo and Juliet have a date at a given time, and each will arrive at the meeting place with a delay between 0 and 1 hour, with all pairs of delays being equally likely. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. What is the probability that they will meet?

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Solution:

$$M = \left\{ (x, y) \mid |x - y| \leq 1/4, 0 \leq x \leq 1, 0 \leq y \leq 1 \right\}$$

